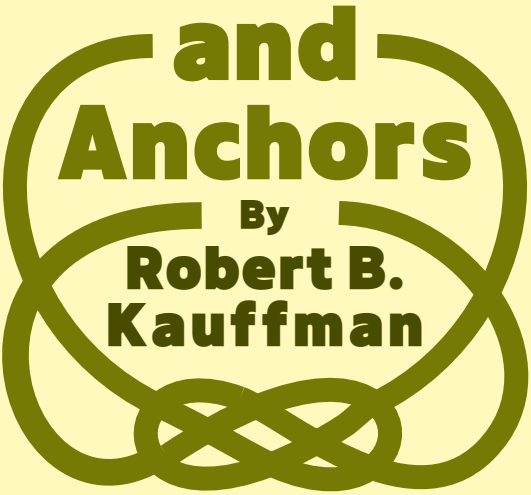
Knots, Hitches, Bends, and Anchors



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Introduction
Principles and Definitions

Knot – The critical element in determining a knot is that when it is tied, it is self-contained and self supporting. It does not need another object or rope to maintain the integrity of the knot (e.g. a hitch does). And, it does not tie two ropes together (i.e. bend).

Also, it should be noted that the term knot is often the generic term used for hitches and bends much like Xerox® is synonymous with copying and used interchangeably with all copying. Similarly, Kleenex® is synonymous with tissue paper and used interchangeably with it. So generic is the use of knot as part of the nomenclature, obvious bends such as the water knot, grapevine knot, or barrel knot are actually called knots. This section recognizes the difference between knots, hitches, and bends. It is structured along these distinctions. However, the term knot is used

generically and interchangeably with hitch and bends.

The term knot is used generically to cover knots, hitches, and bends. It is done so here. Technically, knots, hitches and bends are structurally different and serve different purposes. In addition, the three types of knots indicate three basic knot tying situations. The rope needs an object or another rope to maintain its internal integrity (i.e. hitch). It is fasten two ropes together (i.e. bend). Or it is a self-supporting knot in the rope that creates a loop or serves some other function. Along with principles and definitions, the three types of knots are used as the organizational structure and headings in this section.

Bends – Bends are used to tie two ropes together. The water knot is used to tie the two ends of webbing together to create a sling.

Also, the water knot and double fisherman's knot are really bends.

Hitch – A hitch needs an object or another rope to tie it. Without the object or other rope, it will fall apart. The other object or rope assists in maintaining its structure. Without a tent peg or tree limb, a clove hitch will fall apart. Without another rope, the Prusik will fall apart. Both need an object or another rope to maintain its structure and integrity.

Families – Examination of knots reveals that their internal structure tends to repeat itself in other knots. This suggests that knots can be grouped into families. Also, this helps when visually examining knots because the configurations are the same. The figure eight family of knots is probably the largest family of knots and the it is really a variation of the overhand knot. The same configuration can be found in

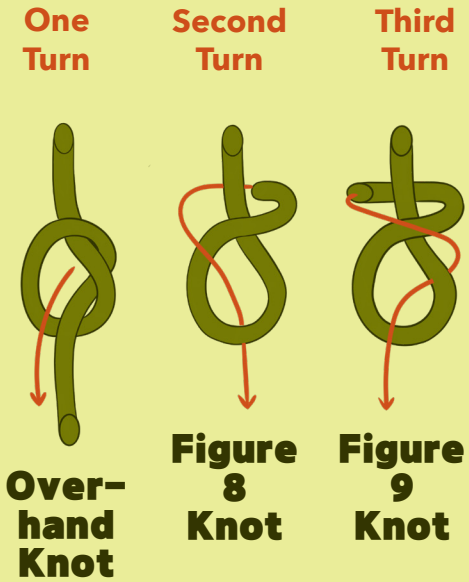
knots, hitches, and bends. The bowline and sheetbend have identical configurations. Two half-hitches and the clove hitch have the same configurations. Two half-hitches are tied around an object. A clove hitch is tied around an object. The fisherman knots are two interlocking overhand knots.

Families can be extended to include similar configurations but where the number of loops or other element is varied in the knot. The single, double, and triple fishermen knots (bends) have the same configuration except for the number of internal loops. The double sheetbend is similar to the sheetbend, except that it has a second internal loop. The barrel or triple fisherman has three internal loops.

Perhaps the most common knot group is the overhand knot (Figure 5.1). Add one more half turn and it becomes the all familiar figure eight knot.

Add a third half-turn and the knot becomes the figure nine knot. So in a very real sense, the figure eight knot is really a variation of the overhand knot. The conclusion is that there are a few basic knots with many variations.

Figure 5.1

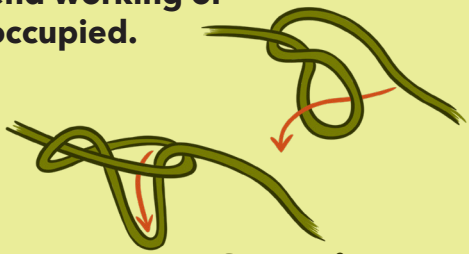


Source: Kauffman, R. (2018). *Swiftwater Rescue Packet.* McHnery, Maryland: garrett College. Unpublished packet.

Figure 5.2 – *Parts of a Rope*

Working End
The end tied off to object being hauled (not shown). It is end working or occupied.

Loop
A turn in the rope that crosses itself



Standing End
All the rope not fastened to the object being hauled.

Bight
A loop that does not cross over itself, usually taken from the center of the rope.

Running End
The end of the rope not tied off and free to use.

Parts of a Rope
Reference: Figure 5.2

The following are some commonly used rope terms. It is useful in orientating and describing the parts of a rope as well as working with a rope. Some of the terminology seemingly overlaps with other terms, such as the difference between a bight and loop. The purpose of labeling the parts is that it helps orient the reader when describing how to tie the knots.

Bight – A bight is a bend in the rope where the line comes back on itself but doesn't cross over itself. It is a loop where the rope doesn't cross over itself. It is usually taken out of the center of the rope.

Loop – In contrast with a bight, a loop is a turn in the rope where the lines cross over each other. When tying a bowline, the first step is to make a loop. The line crosses

over each other and held between the thumb and second finger. The first step in tying a trucker's hitch shown in Figure 5.2 is to create a loop (i.e. actually multiple turns). The second step is to create a bend that is pulled through the loop.

Working End – The working end is attached to the item being rigged or hauled. Think of it as the end of the rope working or currently occupied.

Standing End – The standing part of a rope includes all the rope excluding the working end.

Running End – The running end is the free end of the rope. Sometimes it is called the free end. It is part of the standing end of the rope and it is the section of the rope used to tie a knot, hitch or something else.

Tying Considerations

Some knots are selected because of their strength (e.g. figure-8). Some are chosen because a life might depend on the knot being "bombproof" (e.g. figure-8 on a bight). Some knots are selected because they can be tied quickly and untied easily (e.g. bowline). Others have a specific purpose or functions (e.g. double fisherman, Prusik, water knot).

Function – Many knots are designed for specific uses and functions. A sheetbend and double fisherman are designed to tie two ropes together. A water knot is designed to tie two pieces of webbing together but not two ropes. A bowline is designed to tie a loop in the rope. A clove hitch is designed to tie a rope to a branch or peg.

Strength (Figure 5.3) –

Tie a knot in a rope and it will immediately lose 1/4 to ½ of its strength. A figure eight retains roughly 75% – 80% of its original strength. In contrast, a square knot retains only 43% – 47% of the rope's original strength (Figure 5.3).

The knot creates a stress point where the fibers on the outside of the bend are stretched more and those on the inside of the bend are stretched less. On the inside of the bend, the strands may even become compressed. The differential in stresses can lead to rope failure at this point. A good rule to remember is that when the object around which a rope is bent is at least four times the diameter of the rope (e.g. pulley, branch, a knot, etc), the bend in the rope will be unstressed as a practical matter.

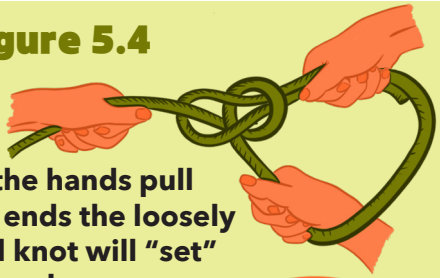
Figure 5.3
Relative Strength of Knots

100%	No Knot
75% - 80%	Figure 8
70% - 80%	Bowline
65% - 70%	Double Fisherman's
60% - 70%	Water Knot
60% - 65%	Fisherman's
60% - 65%	Clove Hitch
60% - 65%	Overhand Knot
60% - 65%	Two Half-hitches
43% - 47%	Square Knot

Setting and Dressed (Figure 5.4) – In tying a knot, the knot is tied loosely.

Setting is the process of tightening the knot so that it becomes dressed. A knot is dressed when it is configured properly. All the parts of the knot are in their proper location and the knot looks as it is pictured in the textbook. This is important because, a properly dressed knot makes for easier inspection and an improperly dressed knot can lose up to 50% of its strength.

Figure 5.4



As the hands pull the ends the loosely tied knot will "set" properly

This knot is not "dressed" properly. Note the kink or unwated twist in the webbing

Ease to Tie/Untie

A knot should be relatively easy to tie. When tying a line around a post or object, the bowline is easier and quicker than a figure-8 follow-through knot is to tie.

Often overlooked is the ability to easily untie a knot. A bowline is easy to untie even after heavy loading. Just bend the "horse" end of the rope and the knot will fall apart. In contrast, once under load, a double fisherman can be virtually impossible to untie. It derives this attribute from its lineage to the overhand knot.

Inspection (Figure 5.5)

It is important to determine that a knot is tied properly and is safe. A properly dressed knot makes it easier to determine if the knot is configured correctly. A figure-8 knot has a unique and distinct configuration. It looks like the figure eight. In a figure-8 follow-through the second rope

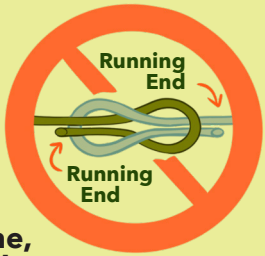
runs parallel to the first rope within the knot. This creates a distinctive pattern. Also, it relates to the knot being properly dressed. that needs to be checked. In the square knot, the key element of the knot that needs to be checked. In the square knot, the two running ends are on the same side of the knot (Figure 5.5). In the "killer square" the running ends are on opposite sides of the knot. Except for this feature, both knots look identical. Unfortunately, the killer square knot will literally fall apart under tension.

Figure 5.5

This is a Square Knot.

"Killer" Square

Except for the running ends, both knots look identical. The "killer" square will come undone, even under a little tension.



Amount of Rope Used

Some knots consume more rope than others. A figure eight on a bight consumes 36% more rope than a bowline. Yet both knots create loops in the end of a rope. A simple experiment was conducted with 8mm cord. With a 2.5" loop, a bowline consumed 11.5 inches of cord. In contrast, a figure eight on a bight consumed 18 inches of cord or 36% more cord than a bowline. This is not insignificant, and at some point, the amount of rope consumed in tying the knot can affect its use.

With its three loops, a triple fisherman's knot consumes more rope than a double fisherman or a single fisherman. Leaving a 1" tail, half of a single fisherman consumed 4.25 inches of the 8mm cord. Half of the double fisherman consumed 5.75 inches or 26% more cord for the additional loop.

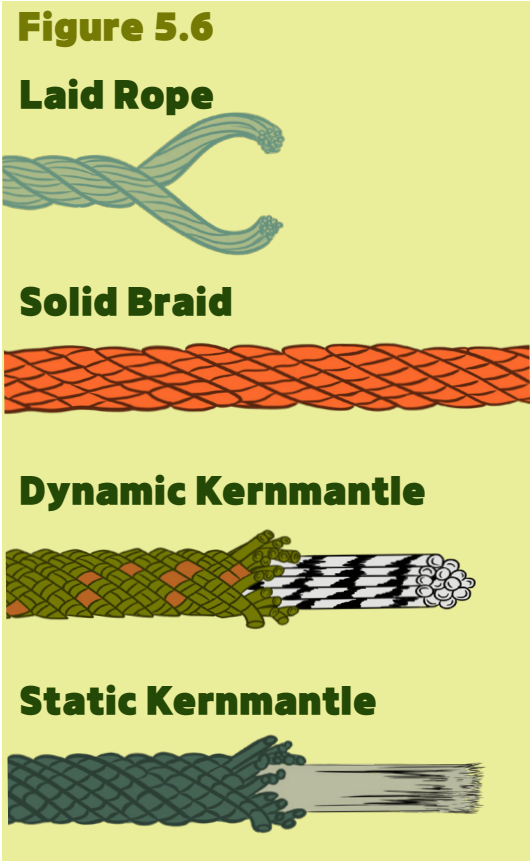
With three turns, half of the barrel or triple fisherman consumed 8.25 inches or 30% more cord than the double fisherman. Since not much strength is gained using a triple fisherman over the double fisherman, the double fisherman can save some rope.

Types of a Rope (Figure 5.6)

Several of the common rope types are presented in Figure 5.6. These are laid, braided and kernmantle. In a laid rope, the strands are usually twisted to be in opposition to each other. This encourages the rope from becoming unraveled. A major disadvantage of the laid rope is that abrasion cuts the outside strands which weaken the rope when that strand move inward and becomes a supporting strand.

The "Kenmantle" is comprised of

the "kern" and the "mantle". The mantle is a woven outer sheathing that protects the kern.



Kenmantle ropes are either dynamic or static. The kern in a dynamic rope consists of twisted

strands of fibers. Dynamic ropes are used in climbing and are designed to stretch. Specifically, they are designed to absorb the fall of a climber. The kern in a static rope consists of parallel unidirectional strands of fibers. The design minimizes stretch. They are used in rescue situations and not in lead climbing or in situations designed to absorb falls.

Ropes are constructed out of nylon, spectra, and polypropylene as well as natural and other materials. Nylon is one of the more common materials. It is strong, flexible, and doesn't float in water. Spectra is much stronger than nylon which is reflected in its cost. Also, it floats. Polypropylene has less strength than nylon. It is inexpensive and it floats in water. In whitewater situations, spectra is often the rope of choice because it is strong and floats.

Chapter One: Knots

The Knot

When tied, a knot is self-contained and self-supporting. It doesn't need an object or another rope to maintain its integrity (e.g. hitch). This may be a fine distinction, but a distinction non-the-less. Traditionally, the term knot is used to include both hitches and bends. To make matters more confusing, many hitches and bends actually incorporate the term knot in their name. The double fisherman's knot and water knot are actually bends. Two families of knots are emphasized in this section, the figure eight and bowline.

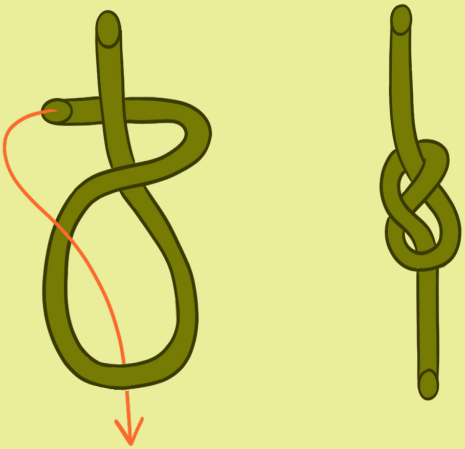
The Figure-Eight Knot Family

Figure Eight Family (Figure 5.7) – Consider the figure eight knot as the basic knot in this group. Actually, it could be argued

that it is really an extension of the overhand knot with one additional turn (see Figure 1). By itself, the figure eight is a stopper knot that is tied at the end of a rope to stop someone who is repelling from dropping off the line at the bottom of the rope.

Other than this, the figure eight by itself has little utility. Other than demonstrating how to tie the knot, this author can't remember ever using the figure eight by itself. Having said this, the figure eight knot is foundational or a base knot for a whole family of knots. The figure eight on a bight, figure eight follow through, or in-line figure eight have utility and considerable usage along with other derivations of the figure eight not noted here.

Figure eight

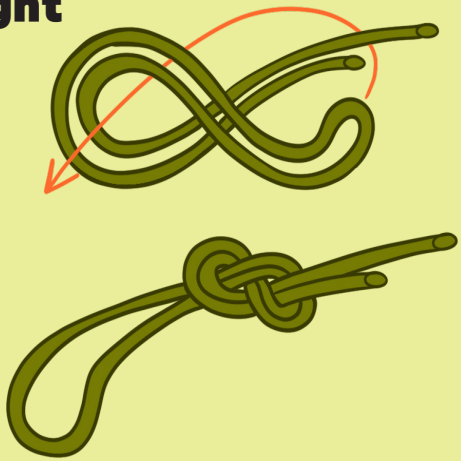


The Figure 8 knot is one of the strongest knots. It can be used as a stopper knot, to tie two ropes together, or to tie a loop in the end of a rope.

Figure 5.7

Figure-eight on a Bight (Figure 5.8) – The figure eight on a bight provides a loop in the end of the rope. The running end of the rope is made into a bight and the bight is tied as if were a simple figure eight. If the person is

Figure eight on a bight



Fold the rope back on itself. Tie a Figure 8 knot with the two strands as you would the Figure 8 pictured above.

Figure 5.8

creating a loop and then clipping it (i.e. carabiner) into something else, this knot is quick to tie and efficient to use. In contrast, if trying to tie a loop around a person or some other fixed object, the figure eight on a bight becomes less practical to use.

Figure eight on a bight follow through

The figure eight on a bight follow-through simply backtracks the figure eight knot in reverse creating a loop or bight in the end of the rope.

Generally, errors result because the back track does not parallel the original

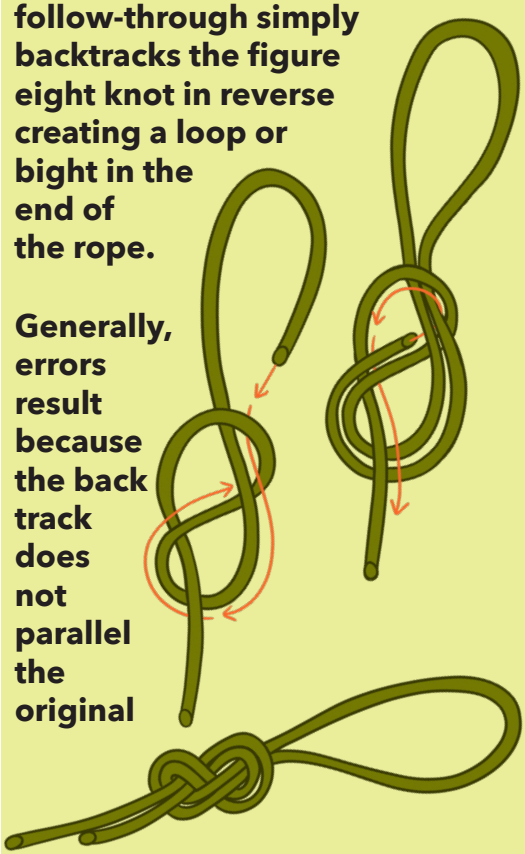


Figure 5.9

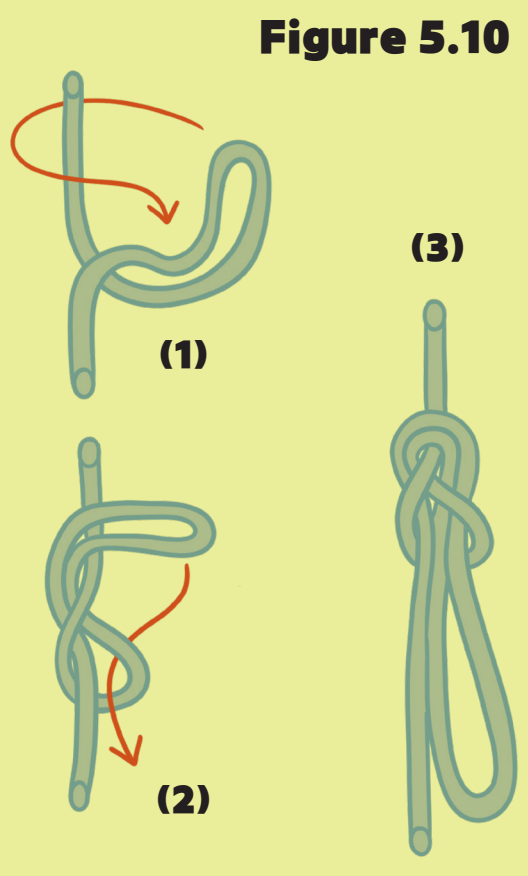
Figure-eight Follow-through (Figure 5.9) –

For climbers, this is the knot used to tie into the harness. It is "bullet proof" in that it won't come undone and it provides maximum protection. It is designed to create a loop in the rope connected to the harness or around the person or other object that is secure and provides maximum protection. It is a staple of climbers. The disadvantage of this knot is that it does take time to tie. Under normal circumstances this is not a problem. In an emergency situation and when time is of the essence, this can become more problematic.

The In-Line Figure-Eight

The in-line figure eight (Figure 5.10) is a figure eight tied onto the middle of the main line. It is a specialized knot. It is not difficult to tie and it is relatively easy to untie. t is superior to tying a overhand knot on a bight which is virtually impossible to untie once

loaded. It forms the base knot in creating a self-equalizing anchor (see next item).



To create the knot:
Make an overhand loop on the rope (1).

Go around the back of the rope, and through the loop (2). Seat Knot (3).

The Directional Figure-Eight Follow-through Knot

The running end of the in-line figure eight is worked back through the loop and then retraced the knot as diagramed. The setup is a self equalizing anchor system where the pull on each anchor is the same. The system can easily incorporate multiple anchor points with the addition of another loop. If there is an abundance of carabiners, they can be used to clip the loops around the anchors to the loop at the bottom of the figure eight.

The system has been used in river rescue and to a lesser extent in climbing. The system can be hooked to multiple D-rings on a raft to extricate it from a pinning. Or it could be used to anchor a haul line to multiple trees used

as anchors where no one tree would serve as a suitable anchor. Actually, it should be used more in climbing since it is truly self-equalizing, particularly when used with carabiners.

To create the knot:
First, create a In-line Figure Eight knot.

Second, loop the running end around the first anchor; lace it through the loop in the Directional Figure Eight; run it around the second anchor; take it back through the loop in the Directional Figure Eight, and then run it around the last anchor.

Third, with the running end from the last anchor tie a Figure Eight Follow Through on the Directional Figure Eight. Finish off the knot with an optional backup knot.

Bowline

The bowline is used to tie a loop in the end of a rope (see Figure

5.12). The advantage of using the bowline is that it can be tied quickly and easily. Also, it consumes less rope to tie than a figure eight follow-through. The knot maintains its integrity under tension. However, it can loosen when placed in continuous tension and compression situations. In the climbing community the bowline has fallen into disfavor for this reason. When a climber falls, they tend to bounce which can loosen the knot. If there is any doubt that the bowline could be placed in a tension and compression situation, back it up.

Untying the Bowline

Even after being placed under extreme tension, the bowline is easily untied. Bend the working end or "neck" forward. This frees the horse collar which can be flipped over the bent working end (i.e. "neck of the horse"). The knot will easily fall apart. As seen in Figure 5.13, Bend the working

end of the rope over the knot (a). This frees the horse collar to be flipped over the working end (b). The knot will become loose and easy to untie.

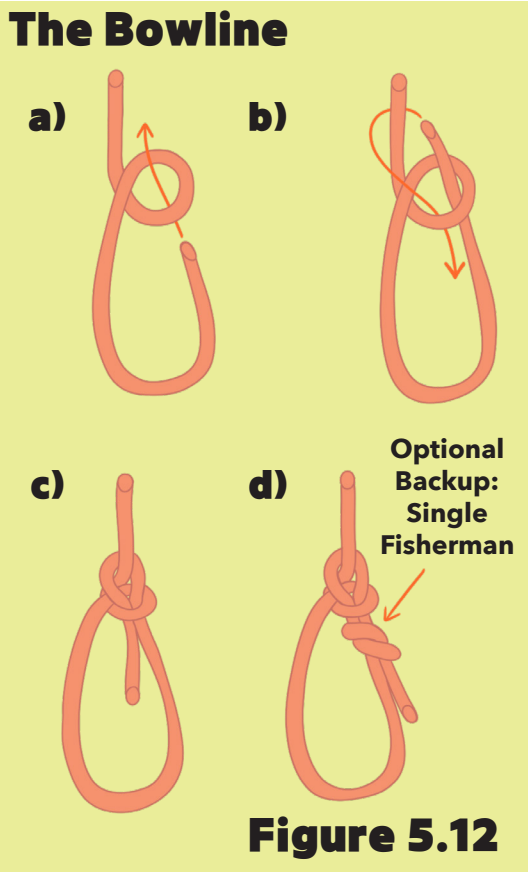
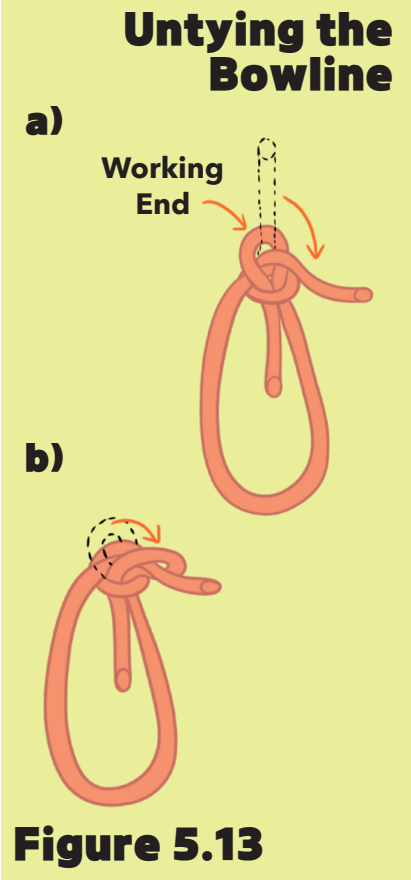


Figure 5.12



A Knot Tying Story: The rabbit comes out of his hole (a); goes around the tree and seeing the fox goes back down into his hole (b). The rabbit grabs his tail and the fox grabs the tree. They both pull (c).

Yosemite Tie-off

If there is a need for a secure bowline, back it off. The Yosemite tie-off is one of several acceptable methods. When tied correctly, the Yosemite tie-off has a figure eight knot look to it. Other knots can be used to back off the bowline. Half a double fisherman's knot can be used (not shown). Half a single fisherman can be used in a pinch, but it may need to be tightened periodically since it tends to be a loose knot.

Chapter Two: Hitches

Prusik – The Prusik is a multi-purpose hitch. It is used in climbing as an ascender and in rescue on hauling systems. In many circles, it is preferred over mechanical devices because it slips at around 900 lbs of tension. This protects the system from fatiguing elsewhere. The Prusik is a hitch because without the main line to maintain its integrity, the knot will fall apart. The Prusik is designed to cinch down on the rope and to kink the rope at an angle to increase its hold. In theory, the Prusick works best when the diameter of the Prusik is significantly smaller than the main line.

In rescue work, this author has used Prusiks with diameters that are close to the main line with little adverse effect. In swiftwater rescue, most mechanical systems use Prusiks in both the hauling system and self-equalizing brake.

The first step in making a Prusik is to make a sling. Tie the two ends of the rope together using a double fisherman's knot. Next, loop the loop around the main line and through itself.

Third is the issue of where the double fisherman knot is located. There are three options. First, the double fisherman can be positioned over the Prusick. In theory, this minimizes the loss of tensile strength due to a knot in the system and in theory, the tensile strength of the Prusik approaches that of the rope without a knot. In practice, it may not make that much difference. The Prusik has two supporting lines. If the rope has a tensile strength of 900 lbs, it has an effective tensile strength of 1,800 lbs. It should be noted that some people find it easier to tie the Prusik by positioning the double fisherman over the knot. In addition, some people find it easier to tie the Prusik by latching onto the knot and passing it through the loops.

The second option is to have the double fisherman located on the side of the sling. As a practical matter, this approach is satisfactory also. Remember, if the Prusik fails (i.e. slips on the rope) at 900 lbs, two 1,800 lb tensile strength cords are not going to fail at the knot even if the knot reduces the strength of the Prusik cord by 50%.

The third option is to locate the double fisherman where the carabiner fastens into the Prusik. This should be avoided since it places undue stress on the system. Often, when tying the knot, the double fisherman is grasped and looped through and around the main line. It is convenient. This approach will tend to line up the knot where the carabiner clips into the sling.

For a Prusik to be effective, it needs to be loosened easily and readjusted along the main line. This is a three-step process of loosening the knot, moving it and re-cinching it. It is important to

loosen the knot first or it will be difficult to move a cinched Prusik.

To loosen or untie the Prusik, pull the "horse collar" over itself. Using the first method, this is the double fisherman. Conceptually, this process is similar to loosening the bowline. Prusiks can easily cinch down on the main line making them difficult to move and readjust.

Trucker's (Rigger's) Hitch – The trucker's hitch is used to fasten boats, construction materials, and other items to roof racks on a car, or to fasten lines to tent pegs that are easily adjustable.

Close inspection of the rig reveals that the hitch is the same configuration as a Z-rig and offers a theoretical 3:1 mechanical advantage. This mechanical advantage makes the trucker's hitch an attractive rig for when the line needs to be taught or when the tension on the line needs to be adjusted. The adjustable lines

to the tent pegs are an example of this re-adjustment. The other feature of the trucker's hitch is that it is easy to untie and when untied, the hitch simply falls apart. A key to having it fall apart easily is to put several twists (i.e. 3-4) in the initial loop.

In situations where the tension needs to be monitored or adjusted, or where there needs to be micro adjustments made in hauling line, the trucker's hitch is ideal for these situations. The author uses it as a staple tool to set up and adjust lines in river rescue including the strainer drill.

Two half-hitches are used to finish off and lock the trucker's hitch into place. If there is any doubt regarding the two half-hitches, use three half-hitches. The half-hitches need to be snug against the loop or it will slide down the line until it becomes snug. If it is anticipated that the system will be readjusted, tie the half-hitches using a bight in the

rope. Pulling on the one side of the bight unties the half-hitches. In addition, extra line can be daisy chained (not shown).

Tensionless Anchor –

Traditionally, the tensionless anchor has been used to anchor ropes around trees. This author has used the tensionless anchor to fasten boats, construction items and other items to be hauled to the roof racks on his car.

It has replaced the trucker's hitch in this use. It works by the rope creating friction around the anchor. It is a hitch because it uses the tree or roof rack to maintain its integrity and structure. In contrast, the anchors depicted in the anchor section use slings to create the anchor. At some point there is considerable overlap between a hitch which requires an object for it to wrap around and an anchor. In the end, it may be a fine distinction.

The literature shows several ways

to finish off the tensionless anchor. The first is to let the free end hang loose (not pictured). This is not recommended. If the tensionless anchor is only under tension, the system will tend to maintain its integrity and not loosen. Regardless, this author doesn't like loose ends and prefers a tie off method. However, if the system is under repeated tension and slack, the anchor will loosen and system will begin to fall apart.

The second method of finishing off the tensionless anchor is tie a figure eight on a bight and fasten the knot to the main line with a carabiner (not shown). This approach is slightly better than using no tie off. It does provide fail safe protection where if the system loosens and begins to unwind, it will eventually lock down on itself. The third approach brings the loose free end around the main line on the first coil and cinching it off. This is why it is important to neatly coil the first two coils. After the first two coils,

the rope can be wrapped in any fashion that is convenient and crossing over itself does not affect the integrity of the anchor.

Munter Hitch –

The munter hitch is a friction device that can be used in place of a rappelling device. The author uses it in place of a clove hitch on a carabiner tied off with two half-hitches (not shown). Either way works. The advantage of using either a tied off munter hitch or clove hitch is that it prevents the carabiner from rotating.

Chapter Three: Bends

Bends tie two ropes together. Three bends are covered, the sheetbend, water knot and double fisherman's knot. A figure eight follow through is not included.

The Sheetbend –

The sheetbend is used to tie two ropes together. Careful inspection of the knot reveals that it has the same configuration as a bowline. Hence, the sheetbend has most of the same attributes as a bowline. It tends to be a loose knot. Under tension and slack, it will tend to become loose and come apart. Like the bowline, it is easy to untie. Simply, break the "horse collar" and the knot falls apart. For additional strength, consider a double sheetbend with two internal loops.

Water Knot –

The water knot is used to tie two pieces of webbing together to

together to form a sling. This is the primary purpose of the knot and no other knot excels like the water knot in this capacity. It is such a stable and strong knot, it does not need to be backed up. Normally, it is not used to tie two ropes together. A sheetbend or figure eight follow through would be preferred.

Literally, the knot is tied as a follow-through overhand knot. Tie an overhand knot loosely in the webbing. With the other end, follow through the knot in reverse. Seat and dress the knot. Avoid kinks. The follow through webbing should parallel the original knot. No backup is needed. The knot maintains its tightness under stress and compression.

Double Fisherman –

Normally, the double fisherman is used to fasten two ends of a rope together to form a loop or sling. Sometimes, it is called the grapevine knot. It is an integral component in creating a sling and

and tying the Prusik. Half of the knot is often used as a backup knot for other knots. The knot cinches on itself making it extremely difficult if not virtually impossible to untie after being loaded. For this reason, it is used in situations where it won't be untied. When tying together two separate lines that will eventually be untied, other knots such as the sheetbend or figure-8 follow-through would be preferred.

The configuration of the single, double, and triple fisherman are essentially the same except for the number of wraps around the rope. The configuration of the single fisherman is that of an overhand knot with the other end of the rope passing through the center of the other overhand knot. A single fisherman's knot has one loop. . A double fisherman's knot has two loops and the barrel or triple fisherman's knot has three loops. A single fisherman's knot tends to be a loose knot and in backup situations, half a double

fisherman is generally preferred. In terms of strength, there is not much difference between a double fisherman's knot and the triple fisherman. However, the triple fisherman's knot consumes more rope. In most situations, the double fisherman is more than adequate in terms of strength and in maintaining its integrity.

Cams Straps

Currently, cam straps have not become a staple in the swiftwater rescue community. However, they are a staple in the rafting and boating communities. They are used to fasten everything to the raft in both paddle craft and oar rigs. For this reason they may be available to use in rescue situations where there are limited resources available. The traditional cam trap is a 1" polypropylene strap with a tensile strength of roughly 1,500 lbs. They come in a variety of lengths ranging from one foot to 20 feet in length.

Cam straps are easy to use and relatively foolproof. Thread the strap into the cam. Pull it taught. The cam is designed to lock down on the strap. To release, simply press the button on the back of the cam.

There are several methods used to store cam straps. Coiling the straps is one of those methods. The advantage of coiling the straps is that they store easily and compactly with other gear. There are no loose ends. They are easy to uncoil. They can be easily tossed to someone who needs a strap. The big disadvantage is that it takes time to coil the straps.

Coiling the straps is not initially intuitive. Two tips to the method include the following. First, thread the webbing behind the cam lever and where the other end of the webbing is attached to the frame. Second, allow sufficient length of the free end so that it can be cinched down later. This may require some trial and error.

Next, the strap shouldhang down (Step 2). Coil the webbing back on itself upward. Coil it toward the cam opening in the frame. When coiled, wrap the free end around the coil and through the cam (Step 3). Cinch it down until it is snug. Store it in the gear bag (Step 4). It is ready for use.

Chapter Four: Anchors

In general, an anchor is a rock, tree or other object to which a line can be firmly attached. A belayer can be considered an anchor. The tensionless anchor discussed in the hitch section is also used as an anchor. The figure-8 follow through makes an excellent self-equalizing anchor. Usually, webbing is used to wrap around the anchor. Three anchors are discussed in this section. They are the simple or single loop, the 3-bight, and the girth hitch.

Simple or Single Loop –

One of two situations occurs. The end result is the same. The webbing is looped around an object or anchor and the water knot is tied connecting the two ends of the webbing together to make a loop. The downside of this approach is having to tie a water knot when tying the sling around the object. It takes time.

Second, a pre-tied sling can be slipped or looped around an anchor which is much quicker. Often, when going around a rock, the extra length of webbing is needed to get around the rock.

The big disadvantage of the single loop is that it can slip down the anchor. This is particularly true on tree trunks and other vertical objects. It is less of a problem going around a rock anchor. The big advantage of the single loop is that it has twice the length of webbing than in the 3-bight approach which can be useful in going around a large rock.

The second version of the simple anchor is much more practical. It can be used on rocks but not trees. Most rescuers will take the sling and loop it around the anchor. This is particularly applicable to wrapping the webbing around a rock or similar anchor. If this doesn't work, the rescuer will often move to the 3-bight approach. It doesn't require retying the sling.

3 Bight –

A water knot is pre-tied in the webbing making a sling. The sling is looped around the anchor and the two ends connected with a carabiner. An advantage of this system is that it is quick and easy to construct. Another advantage of this anchor is that it reduces the load on each of the webbing strands by a fourth of the total load.

It is divided by four supporting lines. As with the simple anchor, a disadvantage of this setup is that it can slip down vertical anchors such as tree trunks. As noted, a disadvantage is that it takes twice the amount of webbing to go around the same anchor as the simple anchor.

Girth Hitch –

In practice, the girth hitch is a practical anchor. It cinches down on the anchor. It can be tied quickly and easily. It is intuitive and simple to use. A pre-tied sling is looped around

An advantage of the girth hitch is that it cinches down on the anchor, making the anchor more effective. A disadvantage of the girth hitch is that because it cinches down on the anchor, it creates a stress point at the hitch. Although this is technically true, it rarely presents an actual problem or system failure.

Summary

The term knot is used generically to cover knots, hitches, and bends. Although the differentiation was used here, it may not make that much difference in practice. The three types of knots indicate three basic knot tying situations. The rope needs another rope or object for it to maintain its internal integrity (i.e. hitch). It fasten two ropes together (i.e. bend). Or it is a self-supporting knot in the rope that creates a loop or serves some other function. Although not a staple in the rescue community cam straps were introduced. In addition, anchors and carabiners were discussed.

Consider knowing how to tie one knot in each of these three situations. The corollary is that sometimes less is more. Consider being able to tie these three knots blindfolded, backwards, or forwards. The point is that knowing how to tie a few knots well is better than trying to tie multiple knots with

overlapping functions poorly or with uncertainty.

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